

Assistive Device for the Speech Impaired

PRAVEEN I.V., KAMALAKKANAN RAVI, BALAMURALI M. AND POONGUZHALI S.

Speech-impaired people have difficulty in communicating with normal people because the hand gestures used by them to communicate their information is not easily understandable; only trained people can understand these. Most expressions and emotions remain un-conveyed, sometimes even misinterpreted. So hand gestures (limited gestures) are not an effective method of communication for the speech-impaired people.

To take care of this issue, an as-

sistive device can be used to enhance the passage of communication. Here, the assistive device consists of a specialised keypad in which each key corresponds to a pre-assigned recorded audio that can be played to convey the message by the speech-impaired person.

Circuit and working

The device consists of transmitter and receiver sections. In the transmitter section (Fig. 1), the keypad matrix is connected to digital input/output (I/O) pins 2-7 of Board1 (Arduino Uno), which is used as the input



device. The microcontroller (MCU) on Board1 scans the key being pressed and sends a 4-bit code representing the key pressed to I/O pins 10 to 13 of

Arduino board.

These four bits are transferred to HT12E encoder for conversion of parallel data into serial data, which is fed into the RF transmitter. It then transmits this serial data at 433MHz using ASK modulation. The 750k resistor determines the oscillator frequency of the transmitter side.

In the receiver section, the code is received using the RF receiver and data bits are decoded if and only if the

extracted address from the transmitted code matches the pre-programmed address in HT12D address pins, thereby avoiding interferences from the commercially-available receivers.

If the decoded address and pre-programmed address matches, LED1 glows, which is connected at pin 17 of IC HT12D. A 33-kilo-ohm resistor (R8) determines the oscillator frequency of the receiver. The decoded output from HT12D is sent to Board2 (Arduino Uno). Arduino board is programmed to generate a binary code, which is sent to IC 74HCT154 decoder to further trigger voice IC APR33A3.

In response to the trigger from the decoder, the voice IC generates corresponding audio messages programmed earlier.

The number of I/O pins on Arduino board is insufficient to drive voice IC APR33A3. So,

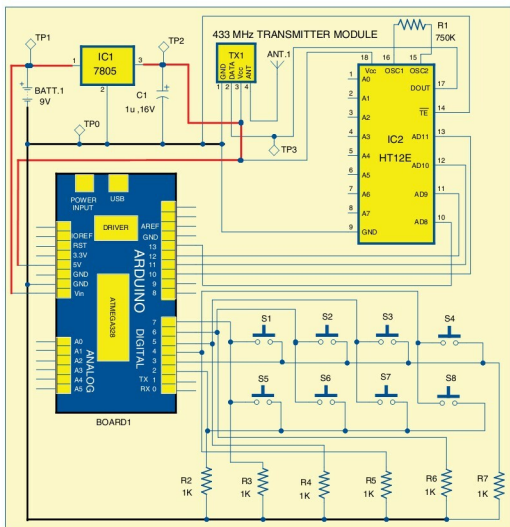


Fig. 1: Circuit of the transmitter section